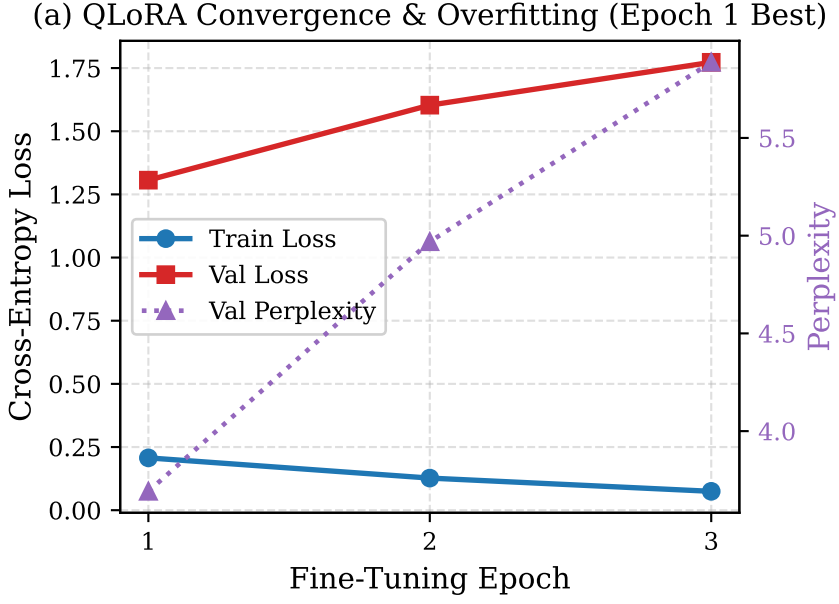
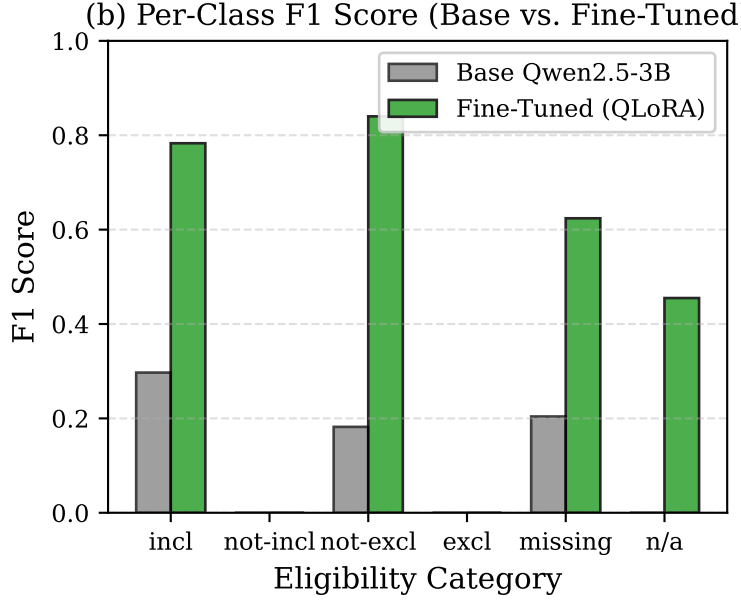

(a) QLoRA Convergence & Overfitting (Epoch 1 Best)
Cross-Entropy Loss
Perplexity
Fine-Tuning Epoch
Train Loss
Val Loss
Val Perplexity
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
4.0
4.5
5.0
5.5
1
2
3


(b) Per-Class F1 Score (Base vs. Fine-Tuned)
F1 Score
Eligibility Category
Base Qwen2.5-3B
Fine-Tuned (QLoRA)
0.0
0.2
0.4
0.6
0.8
1.0
incl
not-incl
not-excl
excl
missing
n/a